INCH-POUND
MIL-DTL-915/58F
22 August 2002
SUPERSEDING
MIL-C-915/58E
29 September 1989

DETAIL SPECIFICATION SHEET

CABLE, POWER ELECTRICAL, 600 VOLTS, FOR OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE MWF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-915 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper, conductor, AWG no. 18, class B stranding, tin coated.
Second - Insulation of synthetic rubber or cross-linked polyethylene. Nominal thickness of 0.020 inch.
Third - Jacket of clear polyamide in accordance with ASTM D 4066, nominal thickness 0.005 inch. Standard identification code applied by method 1.
Fourth - The required number of conductors (see table I) cabled together with a maximum lay of 20 times the pitch diameter of the layer. Cabling sequence to be consecutive, starting with no. 1, from the center outward. Fillers as necessary in order to form a firm, round assembly. Fillers shall be nonfibrous.

Construction (watertight) (continued)

Fifth - Binder tape applied helically with overlap.
Sixth - Arctic type neoprene jacket, black. (See table I for thickness). Cable surface marking required.

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TABLE I. Details.

Type and size	Number of conductors	Cable jacket thickness nominal (inch)	Overall diameter	
			Minimum (inches)	Maximum (inches)
MWF – 7	7	0.060	0.480	0.500
MWF – 10	10	0.065	0.605	0.635
MWF –14	14	0.065	0.605	0.635
MWF – 19	19	0.085	0.710	0.745
MWF – 24	24	0.085	0.800	0.836
MWF – 30	30	0.110	0.905	0.945
MWF - 37	37	0.110	1.005	1.045

EXAMINATION AND TESTS:

Requirements

Basic electrical:

- Conductor resistance - ohms/l000 feet at 25°C, maximum 7.05
- Voltage withstand - volts, root mean square, minimum
  - Conductor to conductor 2000
  - Conductor to water 1000
- Insulation resistance - megohms-1000 feet, minimum
  - Conductor to conductor 300
  - Conductor to water 300

Group A:

- Visual and dimensional ..... No failure
- Hydrostatic (open end) (on every shipped length) –
  - leakage at 1000 lbf/in^2 , in 6 hours, in^3 0

Group B:

Physicals (unaged)

- Insulation (conductor)
  - Synthetic rubber
    - Tensile strength - lbf/in^2, minimum 700
    - Elongation - percent, minimum 350
  - Cross-linked polyethylene
    - Tensile strength - lbf/in^2, minimum 1800
    - Elongation - percent, minimum 350
- Jacket (cable)
  - Tensile strength - lbf/in^2, minimum 1800
  - Elongation - percent, minimum 300

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Set - inch, maximum 3/8
Drip at 95 ± 1°C ..... Zero

Group C:
Physicals (aged)
Insulation (conductor)
Synthetic rubber
Air Oven
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Cross – linked polyethylene
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation- percent of unaged, minimum 80
Permanence of printing (conductor) - cycles, minimum 50
Permanence of printing (jacket) – cycles, minimum 250
Cable filler removability ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, plus all of groups A, B, and C, plus the following:

Cable aging and compatibility (125 ± 5°C) ..... No failure
Cold working (minus 54 ± 2°C) ..... No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
7	1000
10	1000
14	500
19	500
24	500
30	500
37	500

PART OR IDENTIFYING NUMBER (PIN): The PIN consists of the letter M, the general specification number along with the specification sheet, and a dash number representing type and size. Refer to MIL-DTL-915 for an example.

M	**915**	**/58-**	**X-X**
Prefix for Military Specification	Specification Number	Specification Sheet Number	Type and Size

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy - SH
(Project 6145-2317-010)

Review activities:
Army – AR, CR
Navy - CG, EC
DLA - CC

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